

Part I — The committer on its own

Setup: the committer machines

Nineteen machines, each 64 cores and 156 GiB, running three signature verifiers, one coordinator, one sidecar, one load generator, six validator-committers, twelve YugabyteDB tablet servers and three YugabyteDB masters. The six validator-committers and the three masters sit on nine of the twelve database machines, and no machine runs both a validator-committer and a master.

Storage is two 969 GiB XFS volumes per machine, `noatime`, mounted `/data1` and `/data2`. Each database node passes YugabyteDB both volumes as separate data directories (`--fs_data_dirs`, one path per volume) rather than a stripe. The sidecar's ledger uses `/data1` alone.

Transactions carry two read-write operations over 32 B keys, 262 B serialized, in 10,000-transaction blocks, and each transaction slot takes a fresh key, so there are no read-write conflicts. Signatures are ECDSA P-256. Blocks come from the load generator's embedded mock ordering service, which cuts and signs them itself and serves them to the sidecar over the Atomic Broadcast API, so no ordering service is in the path.

A measured point is the highest offered rate that arrived in full, committed within 2 %, kept p99 under 1 s and held a flat in-flight count, confirmed by a 300 s hold on a fresh deployment.

Why this setup

The shape is not the paper's. The published committer deployment is nine validator-committers on nine database nodes, dual 48-core servers with 64 GB. What follows therefore compares deployments, not matched hardware.

Both volumes, not a stripe, so YugabyteDB sees the disk boundary and places data across it itself. This also has to be stated because it is what a wipe must cover: the data directories are siblings of the deployment directory, not children, so clearing the deployment directory leaves the database intact — which silently invalidated every earlier bring-up of the arm in Part II.

ECDSA rather than Ed25519, so the two arms differ only in whether an ordering service is present. This was Ed25519 until the generator stopped being the constraint: `crypto/ecdsa` signs hedged, building an HMAC-SHA-512 DRBG per signature, which a CPU profile at 193,000 tps put at 8.11 % of process CPU against 8.80 % for the elliptic-curve multiplication it feeds — nonce setup costing more than the work it exists to serve. Deriving the nonce directly is worth 2.06× at 32 cores, about 709,000 signatures a second, above the ceilings below. Entropy was never the mechanism (`GetRandom` was 0.54 %), and two earlier explanations — `getrandom` contention, then unspecific allocation — were both wrong.

A conflict-free workload measures the pipeline's ceiling rather than its conflict behaviour, at the cost of a state database and a ledger that grow with every transaction. Conflicts are then introduced deliberately as their own variable.

Queue flatness is part of the definition because a rate can be committed in full out of a growing queue, which reports the queue's drain time as latency.

Throughput and tail latency

Throughput falls with transaction size (Fig. 1), 604,545 tps at one read-write operation to 274,364 at four: 55 % against the published 41 %. The mechanism is not the published one, which is lock contention in the coordinator's dependency graph. Here both lock-wait histograms have a count rate of exactly zero, and the graph's cost is per key instead — about 420 ns each, flat from one key per transaction to eight.

Invalid signatures are close to free but never profitable: 558,364 tps with none, then 517,455, 519,455 and 518,727 at 10 %, 20 % and 30 % — flat within 0.4 %, not the published 10 % *gain*. Driven instead at one common 559,872 tps, all three deliver it while the tail grows with the invalid share (1.1, 5.0 and 7.5 s): rejection is cheap in throughput and expensive in latency.

Double spends do not reproduce, and a deployment setting is why. At the 120-way tablet pre-split this cluster uses, 0 % holds 518,000 tps and *no* rate qualifies at 5 % or above — about 20,000 tps at tens of seconds with every machine under 10 % CPU, so neither capacity nor a queue. At the default split the same shapes hold: 41,273, 30,000 and 40,909 tps at 10 %, 20 % and 30 % with 334 ms to 533 ms p99, flat rather than declining — while costing the conflict-free case 1.8× (518,000 against 294,893). The split is therefore a workload-dependent trade, not a tuning win. Narrowing the graph's chunk width refuted multi-key batching as the mechanism, leaving why 120 tablets collapse unresolved.

What latency costs at a given throughput

At 10,000-transaction blocks the median has a floor near 125 ms that does not fall with the rate, since nothing in a block moves until the block is cut (Fig. 2). At 500 transactions a block the floor is 40 ms, and 380,100 tps holds at 125 ms median and 208 ms p99 — the median large blocks give at 10,000 tps, at thirty-eight times the rate. The top of that curve measures the generator, whose block preparation caps it near 850 blocks a second (425,000 tps at this size; 450,000 offered delivered 432,309), so the committer's own ceiling at 500 transactions is not yet known.

Part II — End to end, with a real ordering service

The second arm puts a real Arma ordering service in the path. The committer tier is **unchanged from Part I** — the same nineteen machines, the same components, the same tuning — so a difference between the two arms is attributable to ordering.

Setup: the ordering machines

Twenty further machines, each 32 cores and 78 GiB, holding four parties of a router, a consenter, an assembler and one

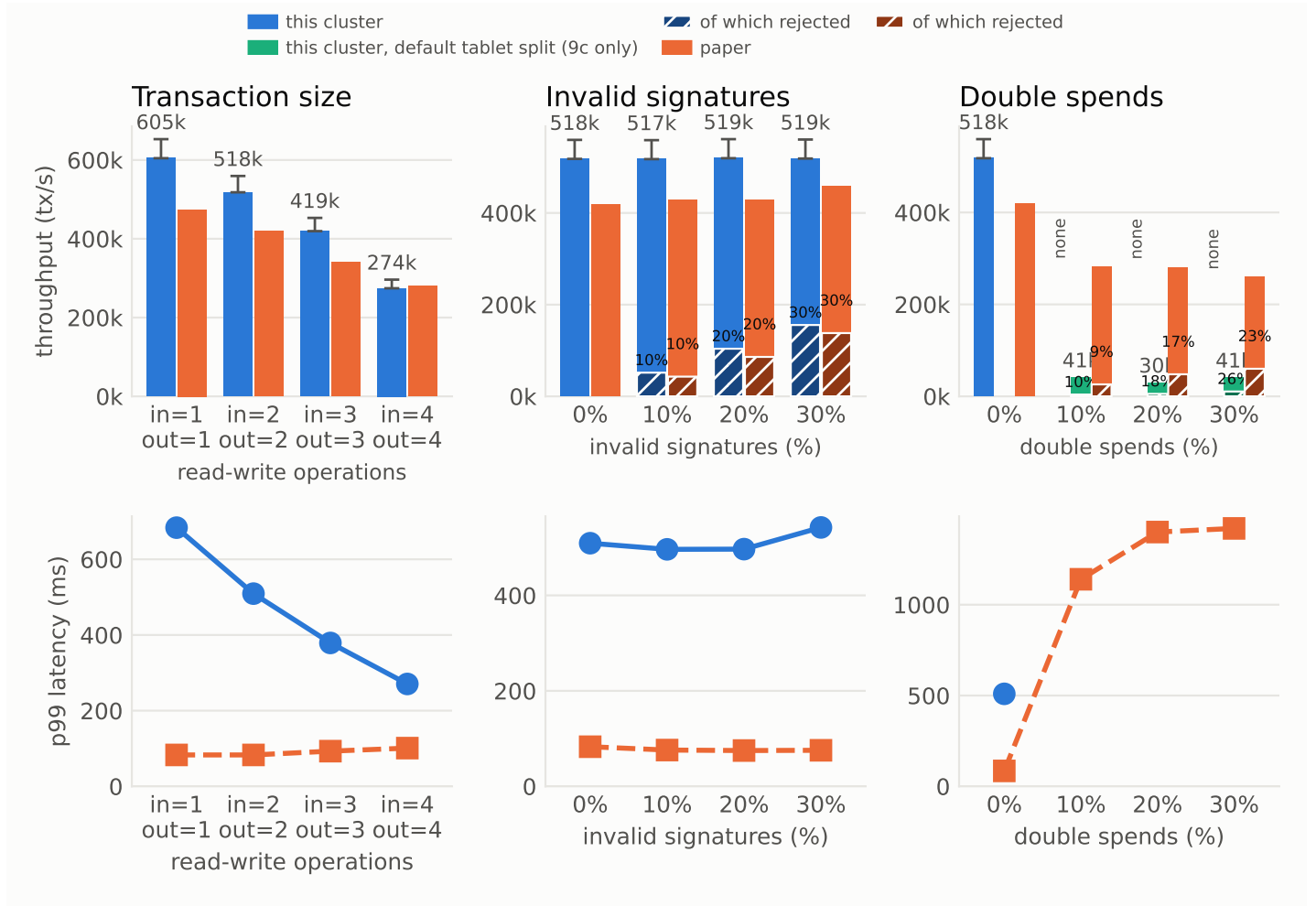


Figure 1: Committer throughput and 99th-percentile latency against transaction size, invalid-signature share and double-spend share, beside the published figures. Each bar is the highest rate held for 300 s with p99 under one second, no queue growth, and the offered rate arriving. Throughput counts transactions *finished*, committed plus rejected; the hatched part of a bar is the rejected share, labelled as a percentage, so the two experiments are compared at matched reject rates rather than at matched labels. Whiskers are the search’s 8% step, one-sided because a knee is a lower bound. *none* marks a condition where no offered rate qualified — the absence is the result. The published series is Figure 9 of the paper: its endpoints as quoted in the text, intermediate points read off the plots at 400 dpi.

batcher per shard. Routers, consenter and assemblers take a machine each, twelve in all; the batchers share the remaining eight. Two configurations are compared:

- **Four shards.** Sixteen batchers, two to a machine, each with a volume to itself (`/data1` and `/data2`).
- **Eight shards.** Thirty-two batchers, four to a machine, two sharing each volume.

Storage is two 485 GiB XFS volumes per machine, mounted as on the committer machines. Routers, consenter and assemblers write to `/data1`; the batchers are distributed across both volumes as above.

Blocks are cut by the batchers, so the block-size knob on this arm is the shared configuration’s `Batching.BatchSize.MaxMessageCount`, not a load-generator setting, and `BatchCreationTimeout` is 500 ms. Signatures are ECDSA P-256, as in Part I, and here the policy derives from a real MSP identity so no other scheme is available. The namespace is created by `fxconfig` before

load is applied, because the transaction that creates it endorses against an MSP rule that the load generator has no material for.

Why this setup

Two batchers to a machine is measured rather than assumed. At two shards the service capped near 158,000 tps *per shard* whatever the batch size or decision interval, while batcher processes sat at 15% to 20% of a 32-core machine — so the constraint is per shard and the spare capacity for a second was already on the box.

Both shard counts are measured because that reading predicts more shards buy throughput, and the two configurations separate the two things a shard needs: a core budget and a disk. Four shards give each co-located batcher its own volume; eight give twice the shards but make two share one.

These machines are half a committer machine — half the cores, half the memory and, per Table 1, half the disk bandwidth — so a ceiling measured on this arm belongs

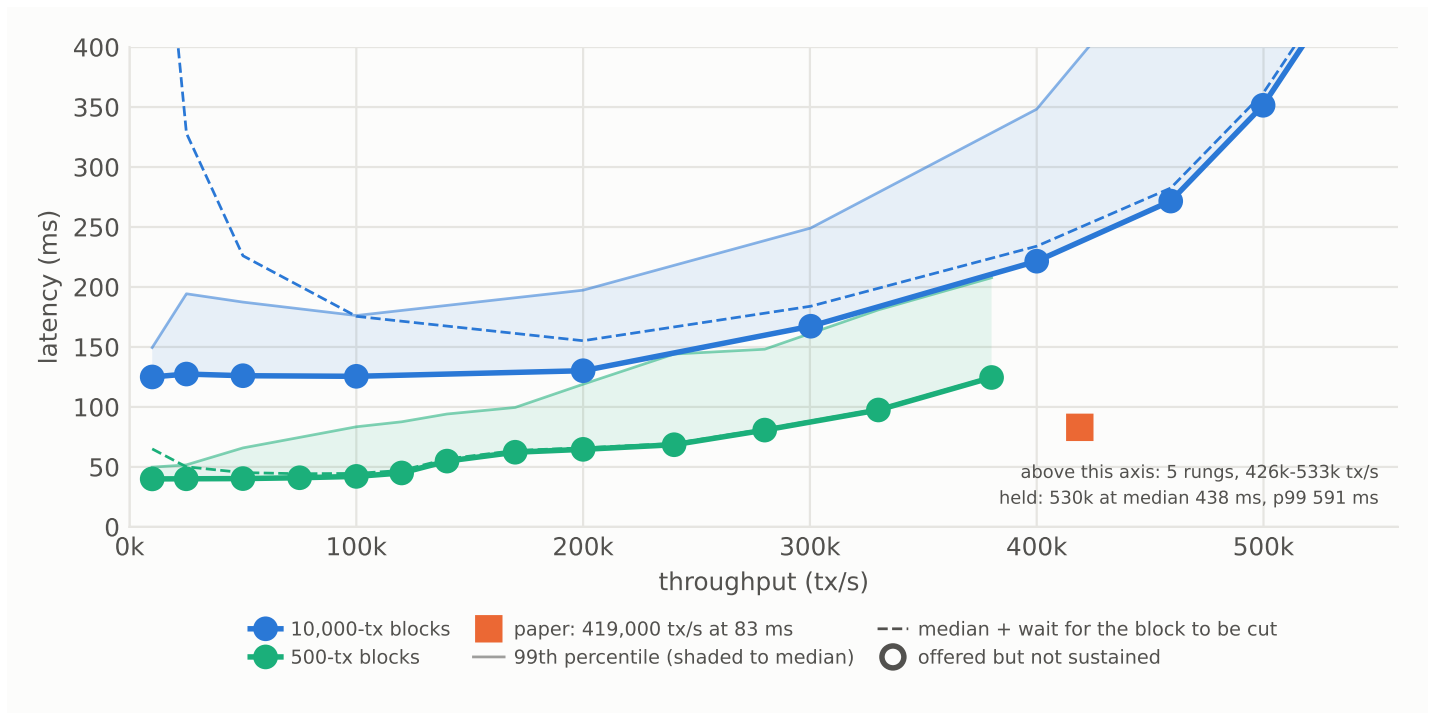


Figure 2: Latency against throughput at two block sizes. The thick line is the median and the thin line above it the 99th percentile, shaded between; the dashed line adds the mean wait for a block to be cut, which the measurement itself excludes. Hollow markers are rates offered but not sustained. The axis is clipped at 400 ms because the region worth reading is 40 ms to 400 ms; what the clip cuts off is named in the corner, and recorded per point in `figures-table.md`.

to these machines before it belongs to ordering.

Shard count and storage

Four shards, each co-located batcher on its own volume, deliver every rate offered up to 300,364 tps with no aborts, a flat queue and p99 473 ms (Fig. ??). That is the top of the ladder, not a knee: the routers were the busiest machines in the deployment and still only 65 % busy, and the median had stopped moving. The configuration's ceiling is therefore above 300,000 tps and not yet known; higher rungs are running, as is the eight-shard ladder. **TODO — the comparison itself.**

Latency *falls* as the rate rises — p99 684 ms at 25,000 tps against 460 ms at 250,000 — which is the 500 ms `BatchCreationTimeout` being the binding constraint at the bottom of the range and not at the top. At low rates a batch waits out the timeout before it is cut; at high rates it fills first. So on this arm the low end of the curve measures the timeout, and only its high end measures the system.

Latency against throughput, ordering included

TODO — running. Two block sizes on whichever shard configuration wins above, with rates low enough that the 500 ms `BatchCreationTimeout` floor is visible. The published work has no end-to-end figure: ordering alone reaches 430,000 tps at four parties and four shards and the committer alone 419,000–474,000, which bracket it.

Transaction size

TODO — queued. One configuration held fixed, varying only transaction size across 300 B, 512 B, 1,024 B, 2,048 B and 4,096 B. The published size sweep is a two-shard measurement falling 413,000 to 27,000 tps over that range, each point within 10 % of 120 MBs^{-1} — so past 300 B it is bandwidth-bound and the rate is simply bytes per second over size. Its shape is the comparable part, not its absolute values, since this arm runs four or eight shards.

Storage characterisation

Table 1 gives `fio 3.35` in the four patterns this deployment produces. The volumes are `virtio` devices reporting neither a model nor a meaningful rotational flag, so the backing media cannot be identified from inside the guest and is characterised by measurement instead.

Two things follow. Ordering machines have half the disk bandwidth as well as half the cores, at almost exactly a factor of two on every bandwidth row, at round figures, repeating to four significant figures across machines and volumes — these are provisioned caps, not device characteristics, so a ceiling on that arm must be read against the cap before it is attributed to ordering. And durability is the expensive direction: making one 4 KiB record durable with an `O_DSYNC` write costs about 400 μs , capping any per-record design near 2,400 operations a second, three orders of magnitude below the rates measured here. The ledger survives that only by batching — it appends a whole block and syncs every hundredth. Syncing alone is cheap by comparison, an `fdatsync`

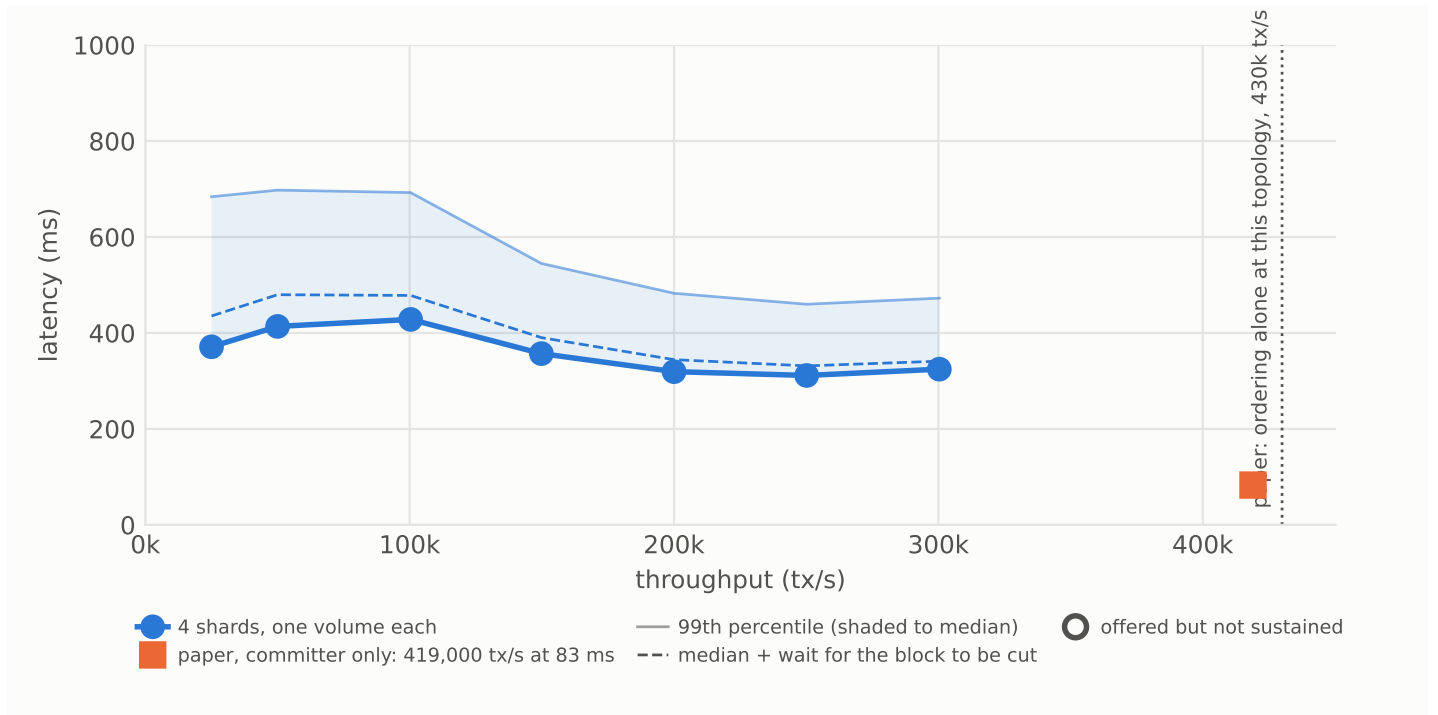


Figure 3: Latency against throughput end to end, with a real ordering service in the path, by shard count and batcher storage layout. Marks as in Fig. 2. The axis is clipped at 1,000ms rather than 400ms, because here the batchers cut the blocks with a 500ms `BatchCreationTimeout` and the median floor sits above 400ms. The dotted line is the published ordering-alone figure for this topology.

Pattern	Depth	Committer	Ordering
1 MiB seq. write	4	1,058 MiB/s	529 MiB/s
4 KiB O_DSYNC write	1	2,400 IOPS	3,249 IOPS
8 KiB rand. read	32	126,000 IOPS	67,689 IOPS
8 KiB rand. write	32	102,000 IOPS	51,165 IOPS

Table 1: `fio 3.35, libaio, direct=1`, 20s per pattern, run on a volume the deployment was not using at the time. The synchronous row is the exception to the pattern: the ordering disks are *faster* there (297µs against 395µs at the median) despite half the bandwidth. Committer figures are the range across a database node and the sidecar. A synchronous engine caps `fio`’s queue depth at 1 while still reporting the depth asked for, so the deep rows use `libaio`.

after a buffered write returning in about 0.08ms: it is the write, not the barrier, that costs.

Threats to validity

The load generator is the instrument and twice became the limit: ECDSA signing, resolved by deriving the nonce directly, and block preparation, which caps the small-block curve in Fig. 2. A knee is resolved to the search’s 8% step, so it is a lower bound, drawn one-sided in Fig. 1. The baseline drifts 10% between days, so only same-day comparisons are used.

The Part I figures were measured with Ed25519 signatures, before the ECDSA path was fast enough to carry them. The setup now specifies ECDSA for both arms; those figures will be re-measured under it, and until they are, Part I and Part II differ in signature scheme as well as in ordering.

One limit is stated in advance of the end-to-end data rather

than after it. That arm measures a ladder on a single deployment, so rate and table fill rise together and the ladder cannot separate them: if its curve bends upward at the top, that is as consistent with a fuller database as with the offered rate. On the committer arm the question was settled by holding the rate fixed while fill grew through one 375s window, where commit latency stayed flat — 142.2ms to 143.0ms across 68% fill growth — but that test belongs to that arm and does not transfer. Until it is repeated here, an upward bend in the end-to-end curve is not attributable to rate. Each rung records the fill it ran against, so the question stays auditable rather than lost.